

AI DOCUMENT & YOUTUBE VIDEO SUMMARIZER

IBM SkillsBuild Internship Final Project Documentation

Project Title:	AI Document & YouTube Video Summarizer
Live Project URL:	https://ai-document-summarizer-58v8.onrender.com/
Internship Program:	IBM SkillsBuild Internship Program
Team Name:	HYDRA
Team Leader:	S Praneeth Reddy
Team Members:	1. Sangireddy Sritharun Reddy 2. Pilla Rajesh 3. Paila Eshwar
Core Stack:	Python (Flask), Groq Llama 3.3 AI, PyPDF, Supadata API, HTML5/CSS3 Glassmorphic UI

1. Executive Summary

In today's fast-paced digital world, users spend considerable time scanning through long articles, dense multi-page PDF documents, and extended YouTube video lectures. The **AI Document & YouTube Video Summarizer** was developed by **Team HYDRA** under the **IBM SkillsBuild Internship Program** to solve this challenge. It provides an efficient, web-based tool that instantly distills long content into clear, concise, structured, bulleted summaries using state-of-the-art AI models.

2. Key Features & Capabilities

- **Multi-Format Input Support:** Process plain text (`.txt`) files, multi-page `.pdf` files, and direct YouTube video URLs.
- **YouTube Transcript Extraction Pipeline:** High-reliability dual-tier retrieval using the Supadata API with direct cloud fallbacks.
- **Ultra-Fast AI Engine:** Powered by Meta's Llama 3.3 70B model via Groq LPUs for near-instantaneous summary generation.
- **Modern Responsive UI:** Engineered with dynamic glassmorphism UI, tabbed navigation, live loading indicators, drag-and-drop targets, and one-click copy functions.
- **Markdown Rendering:** Renders structured headings, bold phrases, and bullet lists using client-side `marked.js`.
- **Production Deployment:** Hosted live on Render cloud platform with full CORS support and HTTPS security.

3. System Architecture & Tech Stack

Component	Technology / Library	Role & Purpose
Backend Framework	Python Flask 3.x	REST API routing, static file serving, backend handling.
AI Inference Engine	Groq SDK (Llama 3.3 70B)	High-speed natural language summarization engine.
PDF Parsing	PyPDF (<code>^pypdf</code>)	Extracting raw readable text from uploaded PDF files.
Transcript API	Supadata API / YouTube Transcript	Automated YouTube transcript fetching engine.
Frontend Interface	HTML5, CSS3, JavaScript (ES6+)	Single Page Application (SPA), glassmorphic styling.
Cloud Hosting	Render Web Service	Production hosting using Gunicorn and automated builds.

4. Backend API Endpoints Specification

POST /summarize/text

- **Payload:** Multipart Form-data ('file')
- **Details:** Ingests text file, decodes text data, and queries Groq AI.

POST /summarize/pdf

- **Payload:** Multipart Form-data ('file')
- **Details:** Parses PDF document content with PyPDF and generates summary.

POST /summarize/youtube

- **Payload:** JSON `{ "url": "" }`
- **Details:** Extracts Video ID, fetches transcript via Supadata API, and summarizes.

5. Team HYDRA - Members & Project Responsibilities

Name	Role	Key Responsibilities
S Praneeth Reddy	Team Leader	Overall System Architecture, Groq Llama AI Integration, Flask Backend Development, Render Cloud Deployment.
Sangireddy Sritharun Reddy	Full Stack Developer	Frontend UI Design (Glassmorphism), CSS animations, marked.js Integration, Single-Page UX.
Pilla Rajesh	Backend Specialist	PyPDF Text Extraction Engine, File Upload Handling, Data Validation Rules.
Paila Eshwar	QA & Integration Engineer	YouTube Transcript Pipeline (Supadata API Integration), Testing, Documentation.

6. Conclusion & IBM SkillsBuild Internship Acknowledgment

The completion of this project marks a successful milestone for **Team HYDRA**. Under the opportunity and mentorship provided by the **IBM SkillsBuild Internship Program**, the team successfully conceptualized, built, and deployed a production-grade AI solution live at <https://ai-document-summarizer-58v8.onrender.com/>.